MECH 191 — Metallurgy

Semester Metallography Portfolio

EXAMET-5-R-600-HD · 50–500× Reflected Light Microscopy
Fall 2026 · Leeward Community College · Instructor: Kai Santos

How to Use This Portfolio

This portfolio contains 15 modules — one per lecture week of MECH 191. Each module is completed during or immediately after class using the shared class specimen kit.

For each module you will: (1) check out the kit, (2) examine assigned specimens on the EXAMET-5 at the listed magnifications, (3) complete the observation table, (4) produce a sketch or photograph of the key microstructural feature, and (5) answer the short-answer questions — then return the kit before leaving class.

Completed modules are kept together and submitted as a single portfolio at the end of Week 16.

Rule	Detail
Kit handling	Return ALL specimens before leaving class. Never clean with abrasives. Handle by edges only — fingerprints etch and obscure microstructure.
Damage	If a specimen is damaged, notify the instructor immediately.
PPE	Safety glasses and gloves are required during all lab activities.
Missed sessions	Live demonstrations are not repeated for absences (excused or unexcused).

Student Name: ____

Student ID: ________________________

Semester / Year: _______________________ Section: _______________________

MODULE 1 · WEEK 1 | Atomic Structure & Bonding

Grain boundaries are the 2-D interfaces between adjacent crystals. Their size, shape, and misorientation angle directly control strength, toughness, and ductility — the Hall-Petch relationship states that smaller grains produce stronger metals.

SPECIMENS	A – 1018 Low-carbon steel (normalized, etched)
MAGNIFICATIONS	50× – overview scan 100× – grain boundary network 100× – grain orientation contrast 500× – inclusions (MnS stringers)
ETCHANT	Nital 2% (2% HNO_3 in ethanol)

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD metallurgical microscope
- Pre-mounted Specimen A (1018 steel, phenolic mount)
- Lint-free lens tissue
- Cotton gloves or finger cots
- Ruler/scale for field-width grain estimates
- Pencil and this portfolio handout

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – overview scan → 100× – grain boundary network → 100× – grain orientation contrast → 500× – inclusions (MnS stringers)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Grain boundary network visible	50×	◻
Equiaxed grain shape confirmed	100×	◻
Grain orientation contrast (light/dark grains)	100×	◻
MnS inclusion stringers (gray, elongated)	500×	◻

STEP 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Sketch or describe the grain boundary network at 100×. Are the grains roughly equiaxed or elongated?
Answer	
Q2	Atomic bonding in steel is metallic. What property of metallic bonding allows grain boundaries to exist without the material cracking?

Answer	
Q3	Estimate the average grain diameter at 100× (field width ≈ 2.3 mm on EXAMET-5). Count grains across the field and divide the width by that count.
Answer	
Q4	If you increased to 500×, what new detail appears that was invisible at 100×?
Answer	

MODULE 2 · WEEK 2 | ## Mechanical Properties

Mechanical properties emerge from microstructure. Pearlite is harder and stronger than ferrite; a finer grain size raises yield strength (Hall-Petch). This module connects what you measure in a test lab to what you see at the microscopic scale.

SPECIMENS	A – 1018 steel (normalized) B – 1045 steel (normalized) C – 1045 steel with Vickers hardness indent
MAGNIFICATIONS	100× – indent identification and measurement 100× – ferrite/pearlite ratio comparison 500× – lamellar pearlite spacing
ETCHANT	Nital 2%

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens A, B, C (phenolic mounts)
- Stage micrometer or calibrated eyepiece graticule
- Calculator (for HV formula)
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – indent identification and measurement → 100× – ferrite/pearlite ratio comparison → 500× – lamellar pearlite spacing

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
White ferrite grains visible (A vs B comparison)	100×	☐
Dark pearlite colonies visible	100×	☐
Square Vickers indent visible on Specimen C	100×	☐
Both diagonals of indent measurable	100×	☐
Lamellar pearlite lamellae resolvable	500×	☐

STEP 2 — Sketch or Photograph

Specimen: _______________ Magnification: ______× Etchant: _______________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Compare Specimens A and B at 100×. Which has more pearlite? Explain why, given 1045 has more carbon than 1018.
Answer	

Q2	Measure the Vickers indent diagonal on Specimen C at 100×. The field width at 100× is ~2.3 mm. Estimate the diagonal in mm, then calculate: $HV = 1.854 \times F / d_{avg}^2$. (Use F = 1 kgf = 9.81 N; d in mm.)
Answer	
Q3	Hall-Petch: $\sigma_y = \sigma_o + k/\sqrt{d}$. If grain size is refined by 50%, what happens to yield strength? (Qualitative answer is fine.)
Answer	
Q4	Which specimen would you expect to have higher tensile strength — and why? Does the microstructure support that prediction?
Answer	

MODULE 3 · WEEK 4 |
Carbon & Alloy Steels

The iron-carbon phase diagram predicts equilibrium phases. Martensite is a non-equilibrium, diffusionless transformation product formed only by rapid quenching. Comparing three steels across the carbon range illustrates how composition controls microstructure — and therefore properties.

SPECIMENS	A – 1018 steel (normalized) B – 1045 steel (normalized) D – 1095 steel (quenched & tempered martensite)
MAGNIFICATIONS	100× – phase field overview 100× – ferrite vs. pearlite distinction 500× – martensite lath/acicular structure
ETCHANT	Nital 2%

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens A, B, D (phenolic mounts)
- Iron-carbon phase diagram reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – phase field overview → 100× – ferrite vs. pearlite distinction → 500× – martensite lath/acicular structure

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Predominantly ferrite + minor pearlite (1018)	100×	☐
Roughly equal ferrite/pearlite (1045)	100×	☐
Dark acicular or lath martensite structure (1095)	500×	☐
Fine carbide particles in martensite matrix (tempered)	500×	☐

STEP 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Draw a simple schematic of the iron-carbon phase diagram and mark where 1018, 1045, and 1095 each fall at room temperature after slow cooling.
Answer	

Q2	Describe the visual difference between pearlite (Specimen B) and tempered martensite (Specimen D) at 500×.
Answer	
Q3	Why does 1095 form martensite when quenched but 1018 does not — even at the same quench rate?
Answer	
Q4	Tempering was applied to Specimen D after quenching. What microstructural change does tempering produce, and why is it visible at 500×?
Answer	

MODULE 4 · WEEK 5 | Stainless Steel & Cast Irons

Carbon can exist as graphite flakes, nodules, or cementite depending on composition and cooling rate. Stainless steel adds Cr to produce an austenitic (FCC) structure stable at room temperature. Shape of graphite is the single most important variable controlling cast iron toughness.

SPECIMENS	E – Gray cast iron F – Ductile (nodular) cast iron G – 304 austenitic stainless steel
MAGNIFICATIONS	50× – graphite distribution (cast irons) 100× – graphite morphology and matrix 100× – twin bands in 304 SS 500× – detail of graphite tips / twin boundaries
ETCHANT	Nital 2% (E, F) | Marble's reagent (G): 10g $CuSO_4$ + 50 mL HCl + 50 mL H_2O

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens E, F, G (phenolic mounts)
- ASTM A247 graphite morphology chart
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – graphite distribution (cast irons) → 100× – graphite morphology and matrix → 100× – twin bands in 304 SS → 500× – detail of graphite tips / twin boundaries

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Graphite flakes with sharp angular tips (gray CI)	40–100×	☐
Rounded/spheroidal graphite nodules (ductile CI)	40–100×	☐
Ferrite/pearlite matrix surrounding graphite	100×	☐
Equiaxed austenite grains (304 SS)	100×	☐
Straight annealing twin bands crossing multiple grains	100×	☐

STEP 2 — Sketch or Photograph

Specimen: _______________ Magnification: ______× Etchant: _______________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Describe the morphological difference between graphite in Specimens E and F. Why does graphite shape matter for toughness?
Answer	

Q2	In Specimen G, identify annealing twins. Why do FCC metals twin readily during annealing while BCC steels generally do not?
Answer	
Q3	Gray cast iron is brittle despite having a ductile ferrite/pearlite matrix. Explain why graphite flake morphology causes this.
Answer	
Q4	A ductile iron component fractured unexpectedly. What would you look for in the microstructure to determine whether nodules were properly formed?
Answer	

MODULE 5 · WEEK 6 | Nonferrous Metals & Phase Diagrams

Nonferrous alloys rely on different hardening mechanisms than steel. Aluminum uses precipitation hardening; brass relies on solid-solution strengthening. Both show grain structures at similar magnifications to steel but with very different twin densities — a direct consequence of stacking fault energy.

SPECIMENS	H – 6061-T6 Aluminum alloy I – 70/30 Brass (Cu-Zn)
MAGNIFICATIONS	50× – grain structure overview 100× – grain boundaries and twins 500× – twin band detail 500× – Mg_2Si precipitates in aluminum
ETCHANT	Keller's reagent (H): 2 mL HF + 3 mL HCl + 5 mL HNO_3 + 190 mL H_2O | Ferric chloride (I): 5g $FeCl_3$ + 50 mL HCl + 100 mL H_2O

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens H, I (H in cold-cure epoxy; I in phenolic)
- Al-Cu binary phase diagram reference
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – grain structure overview → 100× – grain boundaries and twins → 500× – twin band detail → 500× – Mg_2Si precipitates in aluminum

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Equiaxed aluminum grains (6061-T6)	100×	☐
Fine dark Mg_2Si precipitate dots	500×	☐
Very prominent annealing twins in brass (almost every grain)	40–100×	☐
Multiple parallel twin bands per grain (brass)	500×	☐

STEP 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Compare twin density in Specimen I (brass) vs. Specimen G (304 SS, Module 4). Both are FCC. Which has more prominent twins and why? (Hint: stacking fault energy.)
Answer	

Q2	The 6061-T6 designation means solution heat treated and artificially aged. What does aging do to the microstructure, and why does it increase hardness?
Answer	
Q3	At 500× on the aluminum specimen, describe any fine particles observed. What are they likely to be, and how do they strengthen the alloy?
Answer	
Q4	Using the Al-Cu phase diagram concept, explain why solution treatment must precede aging. What happens if the solution treatment step is skipped?
Answer	

MODULE 6 · WEEK 7 | Mechanical Testing

Mechanical tests produce numbers; microstructure explains why. Hardness testing leaves a permanent indent whose geometry encodes the hardness value. Grain size is a quantitative microstructural measurement (ASTM E112) that directly correlates to strength and toughness through the Hall-Petch relationship.

SPECIMENS	A – 1018 steel (normalized) — grain size measurement B – 1045 steel (normalized) — grain size comparison C – 1045 steel — Vickers indent (refined measurement)
MAGNIFICATIONS	100× – grain intercept counting (ASTM E112) 100× – Vickers indent diagonal measurement 100× – grain structure comparison A vs B
ETCHANT	Nital 2%

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD with calibrated eyepiece graticule
- Specimens A, B, C (phenolic mounts)
- Stage micrometer (to calibrate graticule)
- Calculator, pencil, ruler for test-line intercept counting

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – grain intercept counting (ASTM E112) → 100× – Vickers indent diagonal measurement → 100× – grain structure comparison A vs B

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Grain boundaries countable at 100× (intercept method)	100×	☐
Square Vickers indent with both diagonals measurable	100×	☐
Plastic zone around indent (disturbed grain boundaries)	100×	☐
Grain size noticeably different between A and B	100×	☐

STEP 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Measure both diagonals d_1 and d_2 of the Vickers indent. Calculate HV = 1.854 × F / d_avg² (F = 1 kgf = 9.81 N; d in mm). Report your HV result.
Answer	

Q2	Using the ASTM E112 lineal intercept method on Specimen A at 100× (field width ≈ 2.3 mm): draw a test line, count grain boundary intersections, and estimate ASTM Grain Size Number G.
Answer	
Q3	A tensile test on 1045 normalized steel gives UTS ≈ 750 MPa. Using the empirical relation UTS ≈ 3.3 × HV, does your measured HV predict a consistent UTS?
Answer	
Q4	Impact toughness (Charpy) is sensitive to grain size. If the grain size of Specimen B were refined by heat treatment, would toughness increase or decrease — and why?
Answer	

MODULE 7 · WEEK 8 | NDT Methods

NDT methods detect discontinuities indirectly — through magnetic flux leakage, ultrasonic reflection, or surface-penetrant bleed-out. Metallography is the destructive confirmation method that NDT is designed to avoid. This module bridges the two approaches.

SPECIMENS	A – 1018 steel (baseline reference) E – Gray cast iron (surface defect reference) J – Weld cross-section
MAGNIFICATIONS	50× – full-section scan for gross defects 100× – inclusion and pore identification 500× – defect morphology detail
ETCHANT	Nital 2% (A, J) | Nital 2% (E)

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens A, E, J (J in cold-cure epoxy)
- NDT method comparison reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – full-section scan for gross defects → 100× – inclusion and pore identification → 500× – defect morphology detail

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Surface or near-surface void/pore located	40–100×	☐
Elongated inclusion aligned with rolling direction	100×	☐
Weld zones (FZ, HAZ, BM) visible at 50×	50×	☐
Any pores or lack-of-fusion in weld cross-section	40–100×	☐

STEP 2 — Sketch or Photograph

Specimen: _______________ Magnification: ______× Etchant: _______________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	You see a spherical dark void at the specimen surface. Which NDT method — PT, MT, or UT — best detects this in a real part? Why?
Answer	

Q2	An elongated dark inclusion aligned parallel to the rolling direction is visible at 100×. Would MT produce a strong or weak indication on this inclusion? Explain.
Answer	
Q3	In the weld cross-section at 50×, can you identify any pores or lack-of-fusion regions? Describe their location relative to the weld zones.
Answer	
Q4	Metallography destroys the part. Give two situations where destructive metallographic examination is justified over NDT in production or maintenance.
Answer	

MODULE 8 · WEEK 9 | Quality Standards & Defect Recognition

Quality standards require documented, reproducible measurements. ASTM E112 for grain size and ASTM E45 for inclusion rating both require systematic microscopic examination with written records — this module practices that discipline.

SPECIMENS	A – 1018 steel — formal ASTM E112 grain size report
MAGNIFICATIONS	100× – all grain-size measurements 100× – inclusion survey (E45)
ETCHANT	Nital 2%

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD with calibrated graticule
- Specimen A (phenolic mount)
- ASTM E45 inclusion chart comparators (textbook or standard)
- Examination record form (write in this portfolio)
- Calculator, pencil

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – all grain-size measurements → 100× – inclusion survey (E45)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Grain boundaries consistently resolvable (three test lines)	100×	☐
Type A inclusions (sulfide stringers) identified	100×	☐
Type B, C, or D inclusions noted (if present)	100×	☐
Inclusion severity rated Thin or Heavy vs. chart	100×	☐

STEP 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Complete the grain size measurement using ASTM E112 intercept method (three test lines, averaged). Record your result as ASTM Grain Size Number G.
Answer	

Q2	Perform a simplified ASTM E45 inclusion survey at 100×. For each type (A, B, C, D), record Thin or Heavy severity. Present findings in a table.
Answer	
Q3	A material certificate states 'ASTM Grain Size 7 minimum.' Does your specimen meet this requirement? (G7 ≈ 32 μm grain diameter.)
Answer	
Q4	A quality inspector must document a metallographic examination per written procedure. List four data items that must appear in a conforming examination record.
Answer	

MODULE 9 · WEEK 10 | Casting Processes

Castings solidify from the outside in. The microstructure is a map of how heat flowed during solidification: fine chill grains at the mold wall, columnar dendrites growing inward, and equiaxed grains at the center. Controlling this pattern is the foundry's primary quality lever.

SPECIMENS	K – Sand-cast aluminum alloy (full cross-section)
MAGNIFICATIONS	50× – full cross-section (chill / columnar / equiaxed zones) 500× – dendrite arm spacing (DAS) measurement 500× – pore morphology detail
ETCHANT	Keller's reagent: 2 mL HF + 3 mL HCl + 5 mL HNO_3 + 190 mL H_2O

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimen K (cold-cure epoxy mount)
- Stage micrometer or calibrated graticule (for DAS measurement)
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – full cross-section (chill / columnar / equiaxed zones) → 500× – dendrite arm spacing (DAS) measurement → 500× – pore morphology detail

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Fine chill grains at mold surface visible	50×	☐
Elongated columnar grains pointing inward	50×	☐
Coarser equiaxed grains at center	50×	☐
Dendrite arms measurable in columnar zone	500×	☐
Porosity concentrated near center (shrinkage) or surface (gas)	40–100×	☐

STEP 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	At 50× examine the full cross-section. Sketch the three solidification zones (chill, columnar, equiaxed) and label the approximate position of each.

Answer	
Q2	Measure dendrite arm spacing (DAS) at 500× in the columnar zone. Faster cooling produces finer DAS. What does your DAS suggest about the cooling rate?
Answer	
Q3	Locate any porosity in the cross-section. Is it concentrated near the surface or the center? Explain why porosity concentrates there.
Answer	
Q4	A designer wants to maximize tensile strength in this sand casting. Based on the microstructure you observed, suggest two process changes that would improve the result.
Answer	

MODULE 10 · WEEK 11 | Forming Processes

Plastic deformation physically stretches and rotates grains in the direction of metal flow — a direct imprint of the forming process on the microstructure. Annealing restores equiaxed grains by recrystallization. The grain flow left behind in forgings is the source of their superior fatigue resistance.

SPECIMENS	A – 1018 steel (normalized, equiaxed reference) L – 1018 steel (cold-rolled, ~40% reduction)
MAGNIFICATIONS	100× – grain shape comparison A vs. L 100× – aspect ratio measurement 100× – pearlite band alignment with rolling direction
ETCHANT	Nital 2%

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens A, L (phenolic mounts — rolling direction parallel to polished face)
- Ruler for field-width measurement
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – grain shape comparison A vs. L → 100× – aspect ratio measurement → 100× – pearlite band alignment with rolling direction

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Equiaxed grains (Specimen A, aspect ratio ≈ 1:1)	100×	☐
Elongated grains (Specimen L, aspect ratio >3:1)	100×	☐
Pearlite bands aligned with rolling direction (L)	100×	☐
Grain boundaries parallel to rolling direction (L)	100×	☐

STEP 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	At 100×, compare grain shape in Specimens A and L. Estimate the grain aspect ratio (length÷width) in the cold-rolled specimen.
Answer	
Q2	Cold rolling increases hardness and yield strength but reduces ductility. Which microstructural feature you observed explains this property change?

Answer	
Q3	If Specimen L were annealed at 700°C for 1 hour, describe the expected microstructural change. What is the thermodynamic driving force for that change?
Answer	
Q4	Based on the fiber texture visible in the cold-rolled specimen, explain why forgings are preferred over castings for high-fatigue applications.
Answer	

MODULE 11 · WEEK 12 | Machining Processes

Machining removes material by chip formation, but the cutting zone generates heat and stress that can alter near-surface microstructure. Surface finish and subsurface integrity are quality-critical: a ground surface that looks perfect to the eye may carry thermal damage detectable only under the microscope.

SPECIMENS	A – 1018 steel (baseline edge condition) B – 1045 steel (machined edge) H – 6061-T6 aluminum (machined edge comparison)
MAGNIFICATIONS	50× – edge overview 500× – near-surface grain deformation 500× – surface features / preparation scratches
ETCHANT	Nital 2% (A, B) | Keller's reagent (H)

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens A, B, H
- Surface roughness comparator card (optional reference)
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – edge overview → 500× – near-surface grain deformation → 500× – surface features / preparation scratches

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Prepared edge visible and smooth at 50×	50×	☐
Near-surface grain distortion or deformation layer (steel)	500×	☐
Preparation scratches visible at 500×	500×	☐
Comparison of near-surface condition: steel vs. aluminum	500×	☐

STEP 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	At 500×, examine the prepared edge of Specimen A. Does the near-surface grain structure differ from the interior? Describe any deformation layer you observe.
Answer	
Q2	In machining, tool wear increases cutting forces and heat. What microstructural change could this cause in a high-carbon steel workpiece — and how would it appear at 500×?

Answer	
Q3	Surface finish affects fatigue life. Identify one feature visible at 500× on any specimen that would act as a fatigue crack initiation site on a machined surface in service.
Answer	
Q4	Compare the near-surface condition of steel and aluminum specimens. Which shows more preparation-induced deformation? Why might aluminum be more susceptible?
Answer	

MODULE 12 · WEEK 13 | ## Powder Metallurgy

PM parts are made from compacted powder particles sintered below the melting point. The resulting microstructure retains evidence of original particle boundaries and sintering quality — pore size, shape, and distribution reveal whether sintering was adequate.

SPECIMENS	M – PM sintered iron (vacuum-impregnated epoxy mount)
MAGNIFICATIONS	50× – pore distribution overview 100× – point-count porosity estimate 500× – pore morphology (rounded vs. irregular) 100× – ferrite/pearlite matrix (etched)
ETCHANT	Nital 2% (5–10 sec)

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimen M (vacuum-impregnated epoxy — do NOT use vibratory polisher)
- Eyepiece graticule with grid (for point counting)
- Calculator
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – pore distribution overview → 100× – point-count porosity estimate → 500× – pore morphology (rounded vs. irregular) → 100× – ferrite/pearlite matrix (etched)

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Dark rounded pores visible at 50× (unetched)	50×	☐
Pore distribution: uniform or clustered?	100×	☐
Pore morphology: rounded/isolated vs. irregular/ connected	500×	☐
Ferrite/pearlite matrix visible after etching	100×	☐

STEP 2 — Sketch or Photograph

Specimen: _______________ Magnification: ______× Etchant: _______________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	Estimate porosity % at 100× using a simple point-count: count grid points over pores vs. total. Report as % area.
Answer	
Q2	Are pores in Specimen M rounded and isolated, or irregular and interconnected? What does each morphology indicate about sintering quality?

Answer	
Q3	A PM bearing is designed to retain oil in pores. Based on your estimated porosity %, would this part perform well as a self-lubricating bearing? (Target: 15–25% porosity.)
Answer	
Q4	Compare the matrix microstructure of PM iron (etched) to wrought 1018 steel (Specimen A). Are the same phases present? What differences do you see?
Answer	

MODULE 13 · WEEK 14 | Welding Processes

A weld cross-section displays the entire thermal history of the joining process in a single specimen. At low magnification all three zones — fusion zone, heat-affected zone, and unaffected base metal — are visible, each with a distinct microstructure that directly reflects the peak temperature experienced.

SPECIMENS	J – GMAW mild-steel weld cross-section (full bead)
MAGNIFICATIONS	50× – full cross-section all three zones 100× – fusion boundary detail and HAZ grain coarsening 100× – columnar grain orientation in weld metal
ETCHANT	Nital 2% (10–20 sec; macro-etch with 10% Nital optional for gross zone view)

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimen J (cold-cure epoxy — preserves weld geometry)
- Flexible steel rule (for HAZ width measurement)
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – full cross-section all three zones → 100× – fusion boundary detail and HAZ grain coarsening → 100× – columnar grain orientation in weld metal

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Weld metal (fusion zone): columnar grains, darker etch	50×	☐
HAZ: transition zone, grain size gradient visible	40–100×	☐
Base metal: uniform normalized structure	50×	☐
Sharp fusion boundary demarcation	100×	☐
Coarse-grain HAZ adjacent to fusion line	100×	☐

STEP 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	At 50×, sketch the three zones (FZ, HAZ, BM) and label the fusion boundary. Estimate the HAZ width in mm (field width at 50× ≈ 4.6 mm on EXAMET-5).
Answer	
Q2	The fusion zone solidifies as cast metal with epitaxial grain growth from the fusion boundary. Describe the grain orientation pattern you observe in the weld metal.

Answer	
Q3	The HAZ is the weakest zone in a properly made weld. What microstructural feature explains this — and is it visible at 100×?
Answer	
Q4	A poor weld procedure left a lack-of-fusion defect at the sidewall. At what magnification and location in the cross-section would you expect to find it?
Answer	

MODULE 14 · WEEK 15 | Weld Metallurgy & Failure Analysis

Individual HAZ sub-zones each experienced different peak temperatures — and therefore have different microstructures and properties. The CGHAZ is the critical zone for HIC susceptibility. Fracture cross-sections preserve the failure mechanism: a trained eye can read the fracture origin, propagation direction, and mode directly from the microscope.

SPECIMENS	J – Weld cross-section (CGHAZ examination) N – Fracture cross-section (ductile/brittle transition specimen)
MAGNIFICATIONS	100× – CGHAZ grain measurement vs. base metal 100× – fracture zone identification (N) 500× – fracture surface texture detail
ETCHANT	Nital 2% (J) | Nital 2% (N)

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimens J, N (cold-cure epoxy mounts)
- Calibrated graticule (grain diameter comparison)
- Failure analysis reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 100× – CGHAZ grain measurement vs. base metal → 100× – fracture zone identification (N) → 500× – fracture surface texture detail

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
CGHAZ grains visibly larger than base metal (3–5× diameter)	100×	☐
Grain size gradient from fusion line to base metal	100×	☐
Ductile fracture zone: darker etch, fibrous texture	100×	☐
Brittle fracture zone: flatter, granular texture	100×	☐
Transition between ductile and brittle zones located	100×	☐

STEP 2 — Sketch or Photograph

Specimen: ________________ Magnification: ______× Etchant: ________________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1 At 100×, locate and describe the coarse-grain HAZ (CGHAZ) in Specimen J. Measure the approximate grain diameter in the CGHAZ and compare it to base metal grain size from Module 1.

Answer	
Q2	In Specimen N, identify the ductile and brittle fracture zones at 100×. Which zone initiated the fracture? How do you know? (Look for beach marks or the origin point.)
Answer	
Q3	Hot cracking in welds occurs at the solidification temperature range. What microstructural feature makes low-sulfur steel ($S < 0.005\%$) more resistant to hot cracking?
Answer	
Q4	A weld failed by hydrogen-induced cracking (HIC) in the CGHAZ. What microstructural feature makes the CGHAZ particularly susceptible to HIC?
Answer	

MODULE 15 · WEEK 16 | Corrosion & Course Wrap-up

Corrosion is an electrochemical process — what it leaves in the microstructure reveals the mechanism. Uniform corrosion removes material evenly; pitting concentrates at microstructural defects; intergranular attack follows grain boundaries. This final module synthesises all five course units through a single specimen.

SPECIMENS	O – Corroded carbon steel (cross-section through pit / surface layer)
MAGNIFICATIONS	50× – full cross-section overview of corrosion damage 100× – pit morphology and initiation site 500× – corrosion product layer and grain boundary attack
ETCHANT	Do NOT etch — corrosion features visible unetched. Optional: very light Nital to reveal base metal grain structure adjacent to corrosion.

EQUIPMENT & SUPPLIES

- EXAMET-5-R-600-HD
- Specimen O (cold-cure epoxy — preserves corrosion layer)
- Calibrated graticule (for pit depth measurement)
- Corrosion type reference card
- Lens tissue, gloves

EXAMET-5-R-600-HD — Quick Setup for This Module

1. Clean the specimen surface with a lint-free lens tissue using a single swipe (never circular motion).
2. Handle specimen by its edges — fingerprints etch and will obscure the microstructure.
3. Place specimen face-down on the stage. Start at 50×. Focus using the coarse knob.
4. Switch to 100× without moving the stage. Fine-focus only above 100×.
5. Work through magnifications in sequence: 50× – full cross-section overview of corrosion damage → 100× – pit morphology and initiation site → 500× – corrosion product layer and grain boundary attack

Field widths on EXAMET-5: 50× ≈ 4.6 mm · 100× ≈ 2.3 mm · 500× ≈ 0.46 mm

STEP 1 — Observation Table

Feature to Locate	Magnification	Observed?
Corrosion morphology type identified (uniform / pitting / intergranular)	50×	☐
Maximum pit identified and located	40–100×	☐
Pit initiation site (inclusion, grain boundary, phase boundary)	100×	☐
Corrosion product layer visible (dark oxide/scale)	100×	☐
Grain boundary preferential attack (if present)	500×	☐

STEP 2 — Sketch or Photograph

Specimen: _______________ Magnification: ______× Etchant: _______________

(attach photograph or draw field of view at the magnification that best shows the key microstructural feature)

STEP 3 — Short-Answer Questions

Q1	At 50×, describe the corrosion morphology: is it predominantly uniform, pitting, or intergranular? Estimate the maximum pit depth relative to the surrounding surface.

Answer	
Q2	Identify the microstructural feature at which pitting initiated (grain boundary, inclusion, phase boundary, or surface discontinuity). What does this location reveal about the electrochemistry of the corrosion cell?
Answer	
Q3	A corrosion allowance of 3 mm was specified for this carbon steel part in a marine environment. Based on your measured pit depth, estimate remaining service life if the corrosion rate is 0.15 mm/year.
Answer	
Q4	Synthesis question — connect all five units: How does atomic structure (Unit 1) → steel composition (Unit 2) → inspection (Unit 3) → manufacturing process (Unit 4) → joining and corrosion resistance (Unit 5) form a single chain? Use your microscopy observations as evidence.
Answer	

Appendix A — Class Specimen Kit & Field Reference

The kit contains 15 pre-prepared, labeled metallographic specimens. Do not regrind or re-etch. Return all specimens in the labeled foam tray.

Magnification Field Widths — EXAMET-5-R-600-HD

Magnification	Field Width (EXAMET-5)	Ideal for
50×	≈ 8.0 mm	Full cross-section, solidification zones, gross defects
100×	≈ 2.3 mm	Grain counting (ASTM E112), inclusion survey (ASTM E45), indent identification, grain aspect ratio, ferrite/ pearlite ratio
500×	≈ 0.46 mm	Precipitates, lamellar pearlite spacing, martensite lath detail, pore morphology, HAZ sub-zones, fracture texture, grain boundary attack

Specimen Summary

Specimen	Material	Mount	Etchant
A	1018 steel (normalized)	Hot-compression phenolic	Nital 2%
B	1045 steel (normalized)	Hot-compression phenolic	Nital 2%
C	1045 steel — Vickers indent	Hot-compression phenolic	Nital 2%, 3–5 sec
D	1095 steel (Q&T martensite)	Hot-compression phenolic	Nital 2%
E	Gray cast iron	Hot-compression phenolic	Nital 2%
F	Ductile cast iron	Hot-compression phenolic	Nital 2%, 10–15 sec
G	304 stainless steel	Hot-compression phenolic	Marble's reagent
H	6061-T6 aluminum	Cold-cure epoxy	Keller's reagent
I	70/30 Brass	Hot-compression phenolic	Ferric chloride
J	GMAW weld cross-section	Cold-cure epoxy	Nital 2%, 10–20 sec
K	Sand-cast aluminum	Cold-cure epoxy	Keller's reagent
L	1018 steel, cold-rolled 40%	Hot-compression phenolic	Nital 2%
M	PM sintered iron	Vacuum-impregnated epoxy	Nital 2%, 5–10 sec
N	Fracture cross-section	Cold-cure epoxy	Nital 2%
O	Corroded carbon steel	Cold-cure epoxy	Do not etch

Grading Summary

Each module is graded on four components: Observation table completeness, sketch/photograph quality, short-answer accuracy, and kit return/handling. The 15 weekly modules together are worth 75 points; the holistic end-of-semester portfolio review is worth 25 points — totaling 100 points for the semester.

Component	Points per Module	Notes
Observation table	1	Credit for each feature located and checked
Sketch / photograph	1	Labeled with specimen, magnification, etchant
Short-answer questions (4)	2	0.5 pts each; methodology weighted over final number
Kit return & handling	1	On time, no damage, handled by edges
Total per module	5	
15 modules total	75	5 pts × 15 modules
End-of-semester portfolio review	25	Holistic; all 15 modules as a complete document
COURSE TOTAL	100	